



# IndianOil

## **JOURNEY RISK MANAGEMENT (JRM) STUDY**

### **Salem Terminal to SRI VANGALAMMAN FUEL AGENCIES**

#### **Objective of the JRM Report**

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### **Regulatory Compliance**

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### **Risk Management Strategy**

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

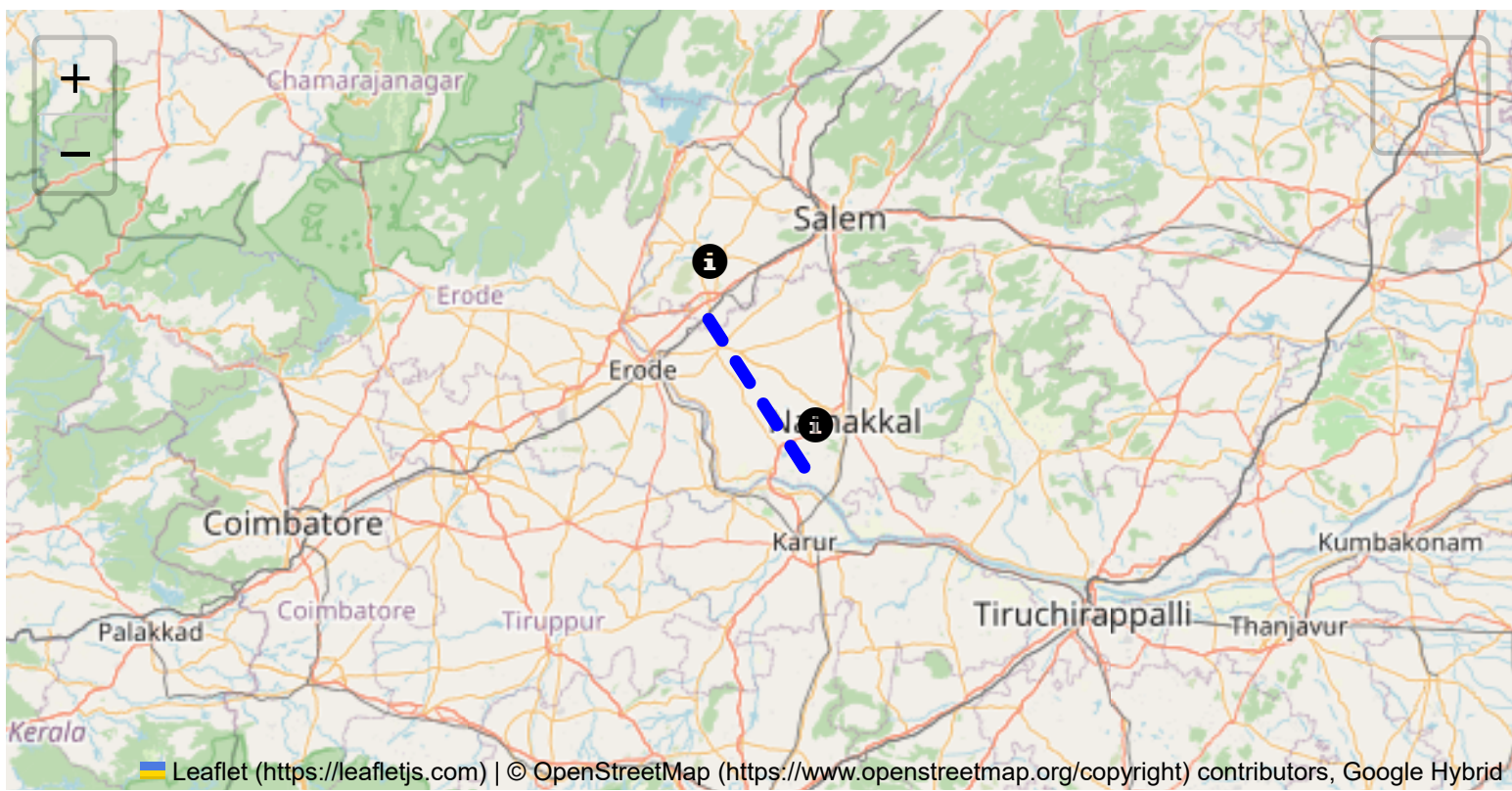
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## **Journey Risk Mitigation**

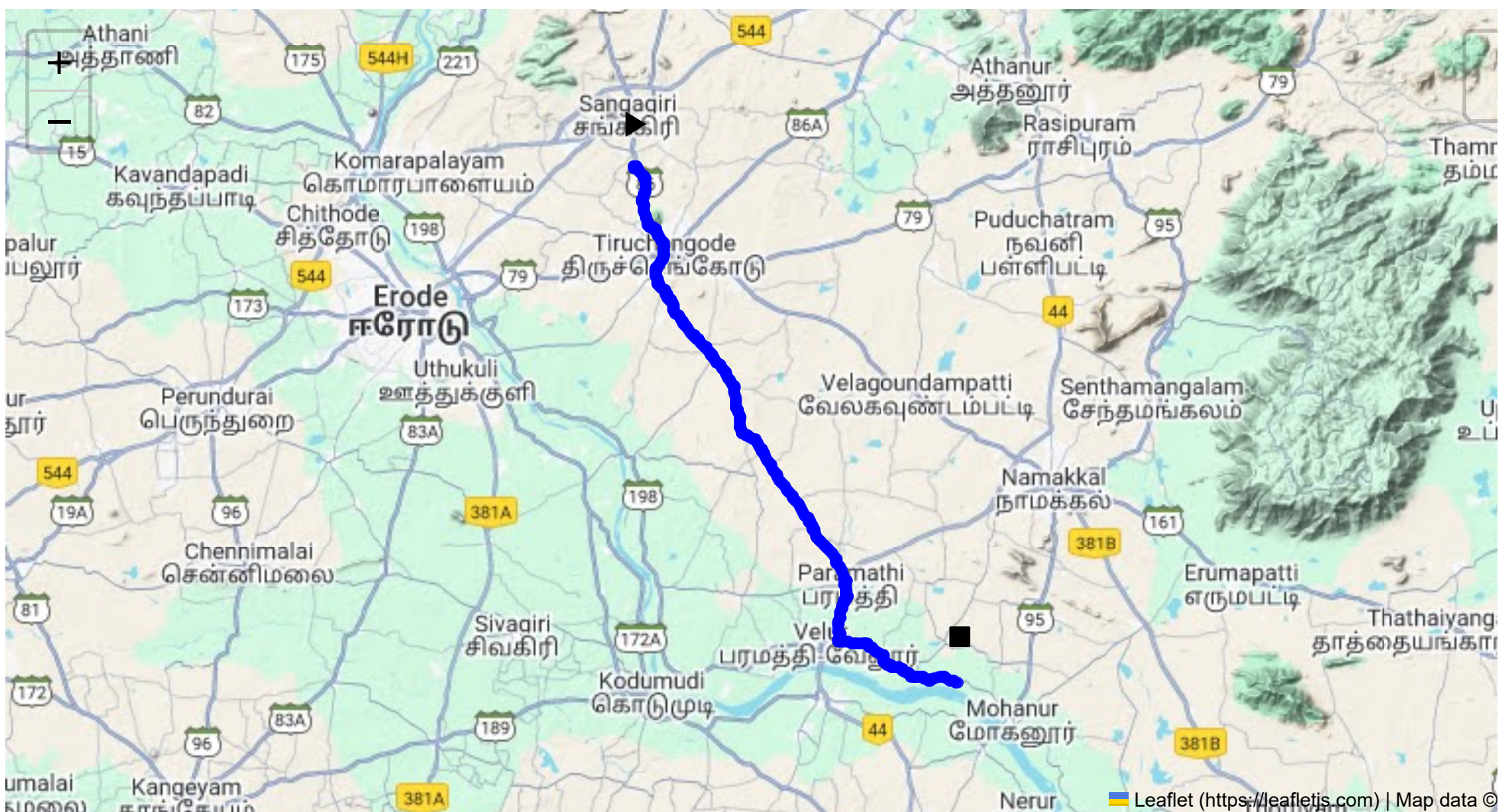
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



**Route Summary:**  
**Total Distance: 52.99 km**  
**Estimated Duration: 1.2 hours**  
**Adjusted Duration (Heavy Vehicle): 1.5 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.086257, 78.099864)**



## Welcome to the Journey Risk Management Study

### Route Overview

The route from CVQF+23W, Sangagiri to Mohanur, Tamil Nadu, covers approximately 52.99 kilometers, with an estimated travel time of 1.18 hours for heavy vehicles carrying hazardous materials. This journey involves traversing NH44 before joining the local roadways towards Mohanur, passing through semi-urban and rural landscapes.

## Weather Conditions and Hazards

Tamil Nadu generally has a tropical climate, with temperatures ranging between 20°C to 37°C. The region experiences the northeast monsoon between October and December, which could lead to heavy rains and occasional flooding, potentially impacting road conditions and visibility. During these months, drivers should be particularly cautious.

## Traffic Patterns

The major highways, including NH44, tend to experience moderate to heavy traffic, especially during peak hours (8:00 AM to 10:00 AM and 5:00 PM to 7:00 PM). Congestion can be expected near Sangagiri town and the approach to Mohanur due to local traffic, markets, and school zones.

## Road Quality and Infrastructure

NH44 is a well-maintained national highway with good infrastructure, including lanes for heavy vehicles. However, local roads leading to Mohanur might have potholes and narrower lanes. Roadwork or maintenance activities could also be present, thus requiring attentive driving.

## Alternative Routes for Emergencies

In case of emergencies, consider using state highways SH85 or SH206 as alternative routes. However, these alternatives may not support heavy vehicle traffic as well as NH44 and should only be used if absolutely necessary.

## Hazardous Material Regulations

Transport of hazardous materials in Tamil Nadu is subject to strict regulations, including proper labeling, safe storage within vehicles, and adherence to designated routes. Drivers must ensure compliance with the Petroleum and Explosives Safety Organization (PESO) guidelines and other relevant regulations.

## Historical Incidents

There are occasional reports of incidents involving heavy vehicles in the region, often attributed to speeding, poor road conditions, or adverse weather. Ensuring mechanical integrity of vehicles and adherence to speed limits can mitigate risks.

## Environmental Considerations

The route passes through some agricultural areas and small water bodies. Spillages of hazardous materials in these regions could pose a significant threat to local ecosystems. It is crucial to follow environmental safety protocols to minimize risks.

## Communication Coverage

The route has decent mobile network coverage, though certain rural segments may experience connectivity issues, particularly for data services. Drivers should verify their cell phone's network

compatibility and travel with backup communication devices if necessary.

## Emergency Response Times

Response times may vary based on proximity to urban centers. Sangagiri has better access to emergency services, with estimated response times around 20-30 minutes. Rural areas near Mohanur may see longer response times, averaging 45-60 minutes.

## Overall Risk Assessment

The route has manageable risks if proper precautions are taken. Weather and road conditions may affect travel times and safety. Adherence to traffic regulations, especially for hazardous materials, and remaining alert to environmental and communication challenges are key to ensuring a safe journey. Regular vehicle checks and updated emergency plans are recommended to address potential incidents effectively.

### Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 2  | Turn       | Medium     | 11.43956, 77.87340 | 30 KM/Hr    |
| 3  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 4  | Blind Spot | Blind Spot | 11.44029, 77.87544 | 10 KM/Hr    |
| 5  | Turn       | High       | 11.37868, 77.89498 | 15 KM/Hr    |
| 6  | Turn       | High       | 11.37850, 77.89453 | 15 KM/Hr    |
| 7  | Turn       | Medium     | 11.16764, 78.01565 | 30 KM/Hr    |
| 8  | Turn       | High       | 11.15573, 78.02176 | 15 KM/Hr    |
| 9  | Turn       | High       | 11.14545, 78.02208 | 15 KM/Hr    |
| 10 | Turn       | High       | 11.14480, 78.02291 | 15 KM/Hr    |
| 11 | Turn       | High       | 11.11887, 78.01758 | 15 KM/Hr    |
| 12 | Turn       | High       | 11.11883, 78.01771 | 15 KM/Hr    |
| 13 | Blind Spot | Blind Spot | 11.11397, 78.01722 | 10 KM/Hr    |
| 14 | Turn       | Medium     | 11.10418, 78.04796 | 30 KM/Hr    |

## Emergency Locations

|    | type     | name                             | coordinates               | speed_limit | risk_level |
|----|----------|----------------------------------|---------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital | 11.3903328,<br>77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331,<br>77.8931963 | 30 km/h     | Medium     |
| 5  | hospital | T.C.A Hospital Tiruchengode      | 11.3791885,<br>77.8965774 | 30 km/h     | Medium     |
| 6  | hospital | Soorya Multispecialty Hospital   | 11.3786429,<br>77.8931912 | 30 km/h     | Medium     |
| 7  | clinic   | Kongu Nursing Home               | 11.3783065,<br>77.8961134 | 30 km/h     | Medium     |
| 8  | hospital | Tiruchengode Government Hospital | 11.37645, 77.89426        | 30 km/h     | Medium     |
| 9  | hospital | Krishna Hospital, Namakkal       | 11.3754811,<br>77.8931817 | 30 km/h     | Medium     |
| 10 | hospital | Tirukumaran Hospitals            | 11.3782118,<br>77.8914933 | 30 km/h     | Medium     |
| 13 | police   | கரட்டுப்பாளையம் காவல்<br>நிலையம் | 11.357586,<br>77.8922539  | 30 km/h     | Medium     |
| 14 | hospital | Goverment Hospital, Nallur       | 11.2679698,<br>77.946863  | 30 km/h     | Medium     |
| 16 | hospital | Goverment Hospital               | 11.156647,<br>78.0213586  | 30 km/h     | Medium     |

## Crowded Spots

|    | type        | name                                   | coordinates               | speed_limit | risk_level |
|----|-------------|----------------------------------------|---------------------------|-------------|------------|
| 2  | school      | அரசு ஆண்கள் மேல்நிலைப்<br>பள்ளி        | 11.3850439,<br>77.8948279 | 30 km/h     | Medium     |
| 3  | school      | அரசு பெண்கள் மேல்நிலைப்<br>பள்ளி       | 11.3844247,<br>77.8949053 | 30 km/h     | Medium     |
| 4  | marketplace | திருச்செங்கோடு தினசரி<br>காய்கறி சந்தை | 11.3833608,<br>77.8970145 | 30 km/h     | Medium     |
| 11 | school      | KSR Educational institution            | 11.3772013,<br>77.8908807 | 30 km/h     | Medium     |

|    | type   | name                                                           | coordinates               | speed_limit | risk_level |
|----|--------|----------------------------------------------------------------|---------------------------|-------------|------------|
| 12 | school | MDV School                                                     | 11.3719843,<br>77.8915244 | 30 km/h     | Medium     |
| 15 | school | Sivabakkiam Muthusamy Matriculation<br>Higher Secondary School | 11.1788177,<br>78.0061696 | 30 km/h     | Medium     |
| 17 | school | Goverment Higher Secondary School                              | 11.1511724,<br>78.0200723 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.38395, 77.89480



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.37868, 77.89498



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.37850, 77.89453



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.16764, 78.01565



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.15573, 78.02176



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.14545, 78.02208



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.14480, 78.02291



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.11887, 78.01758



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.11883, 78.01771



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.11397, 78.01722



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.10418, 78.04796

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